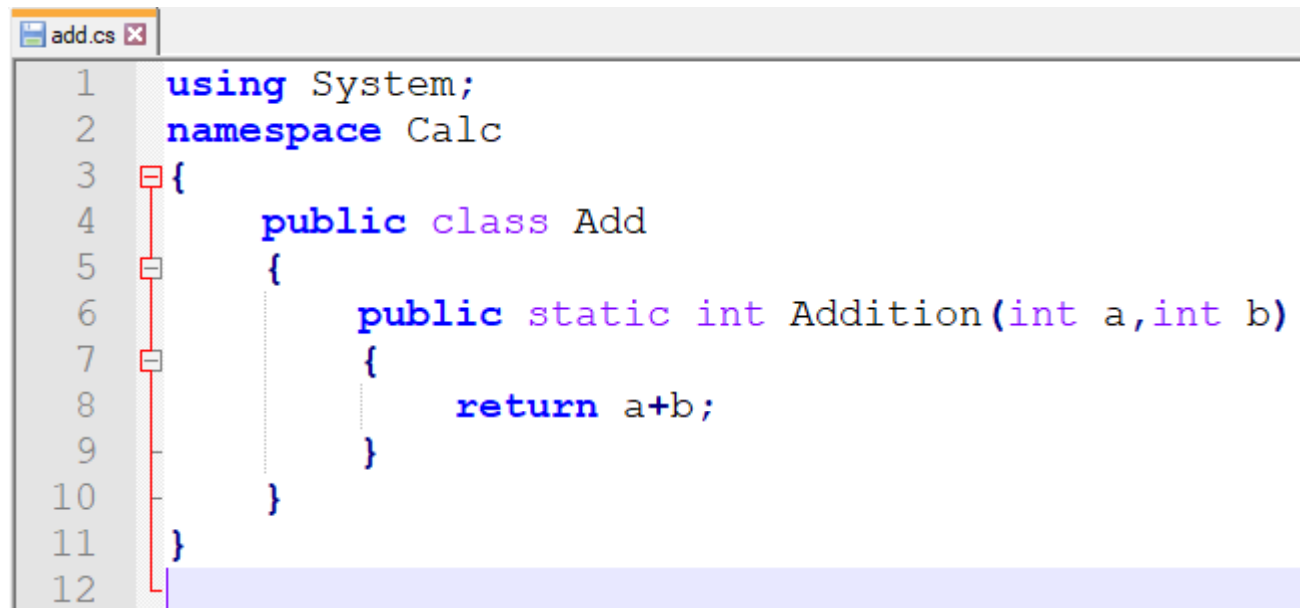


Multifile Assembly, Strongly Named Assembly

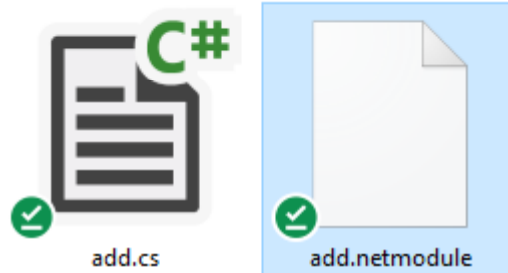
Ex. 1



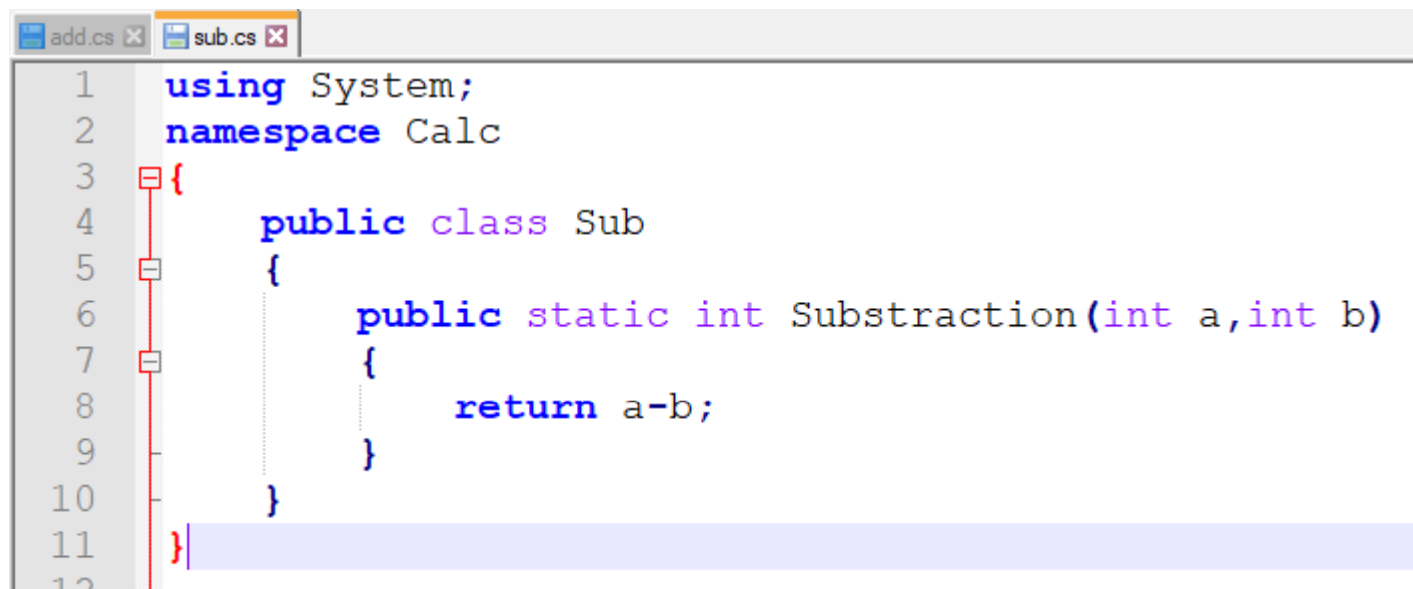
```
1 using System;
2 namespace Calc
3 {
4     public class Add
5     {
6         public static int Addition(int a,int b)
7         {
8             return a+b;
9         }
10    }
11 }
12
```

Ex. 1

```
E:\GoogleDrive\Personal\BTech\WAD\2022-23\Labs\Lab3\ex1>csc /t:module add.cs  
Microsoft (R) Visual C# Compiler version 3.4.1-beta4-19614-01 (16504609)  
Copyright (C) Microsoft Corporation. All rights reserved.
```



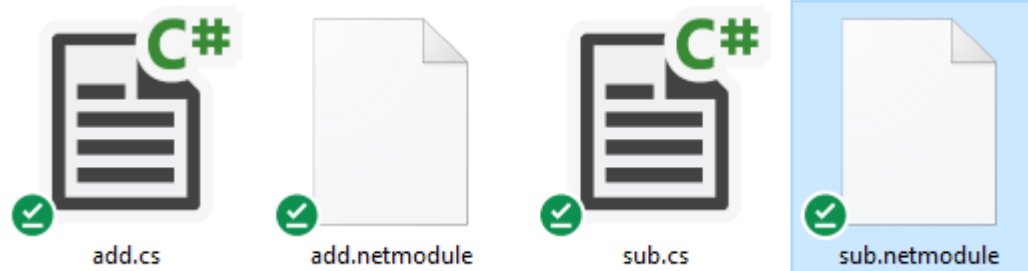
Ex. 1



```
1 using System;
2 namespace Calc
3 {
4     public class Sub
5     {
6         public static int Subtraction(int a,int b)
7         {
8             return a-b;
9         }
10    }
11 }
```

Ex. 1

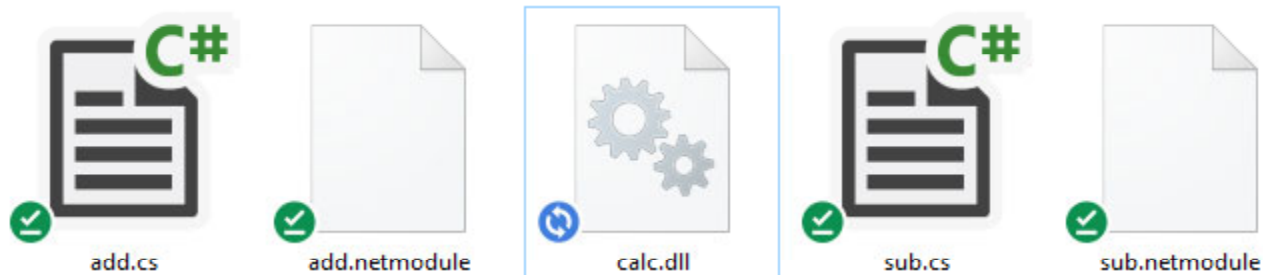
```
E:\GoogleDrive\Personal\BTech\WAD\2022-23\Labs\Lab3\ex1>csc /t:module sub.cs  
Microsoft (R) Visual C# Compiler version 3.4.1-beta4-19614-01 (16504609)  
Copyright (C) Microsoft Corporation. All rights reserved.
```



Ex. 1

```
Developer Command Prompt for VS 2019

E:\GoogleDrive\Personal\BTech\WAD\2022-23\Labs\Lab3\ex1>csc /t:library /out:calc.dll /addmodule:add.netmodule /addmodule:sub.netmodule
Microsoft (R) Visual C# Compiler version 3.4.1-beta4-19614-01 (16504609)
Copyright (C) Microsoft Corporation. All rights reserved.
```

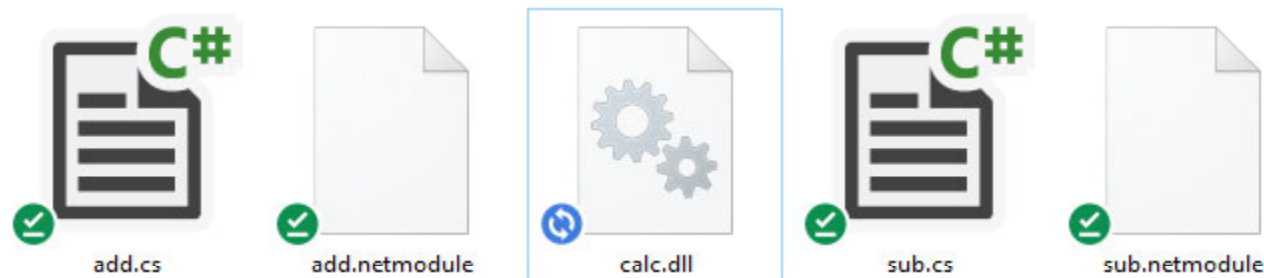


Ex. 1

```
Developer Command Prompt for VS 2019

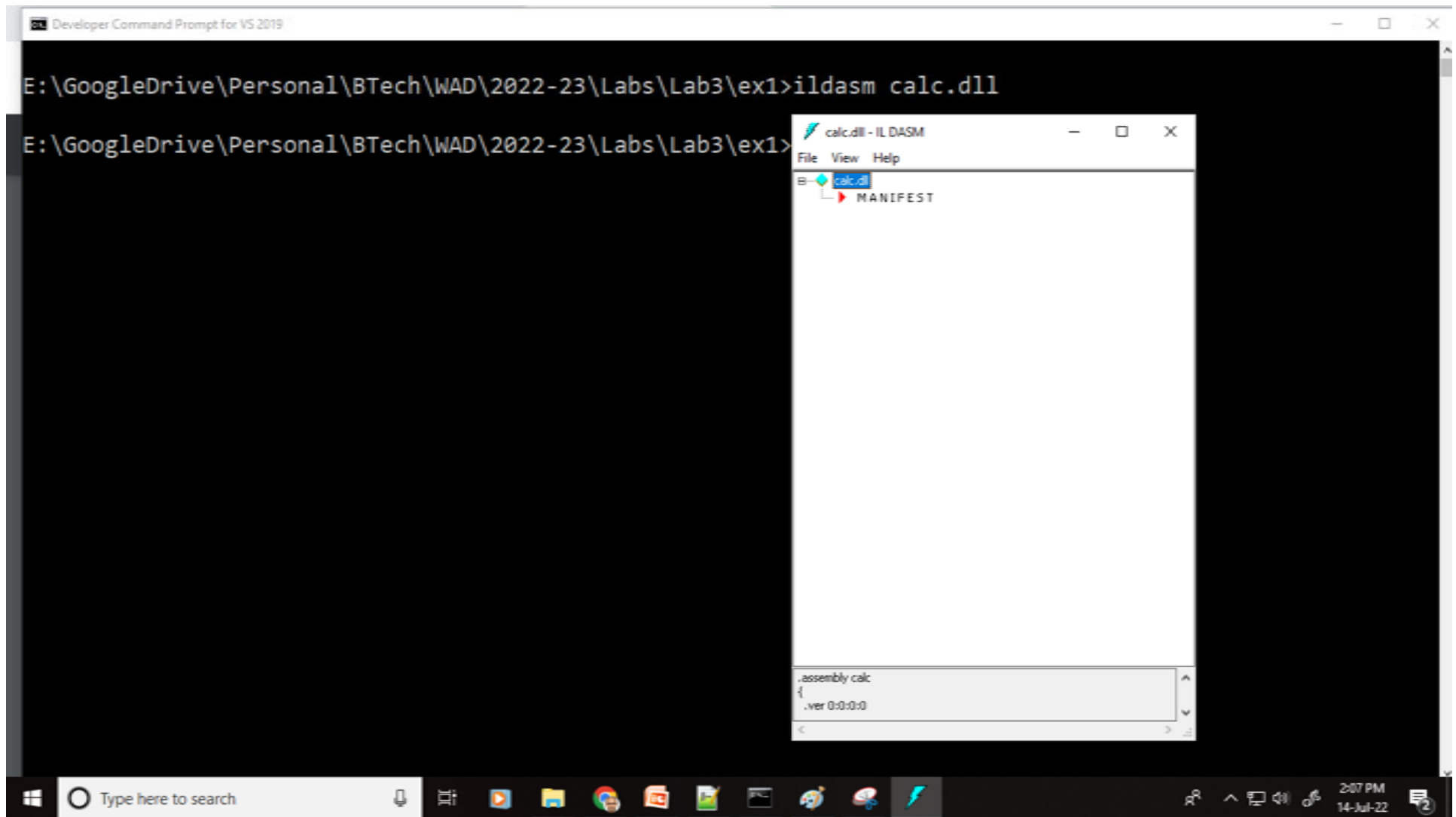
E:\GoogleDrive\Personal\BTech\WAD\2022-23\Labs\Lab3\ex1>al /t:library /out:calc.dll add.netmodule sub.netmodule
Microsoft (R) Assembly Linker version 14.8.3928.0
Copyright (C) Microsoft Corporation. All rights reserved.

E:\GoogleDrive\Personal\BTech\WAD\2022-23\Labs\Lab3\ex1>_
```



Is the manifest standalone or attached to one of the assembly?

Where is the IL, Metadata?



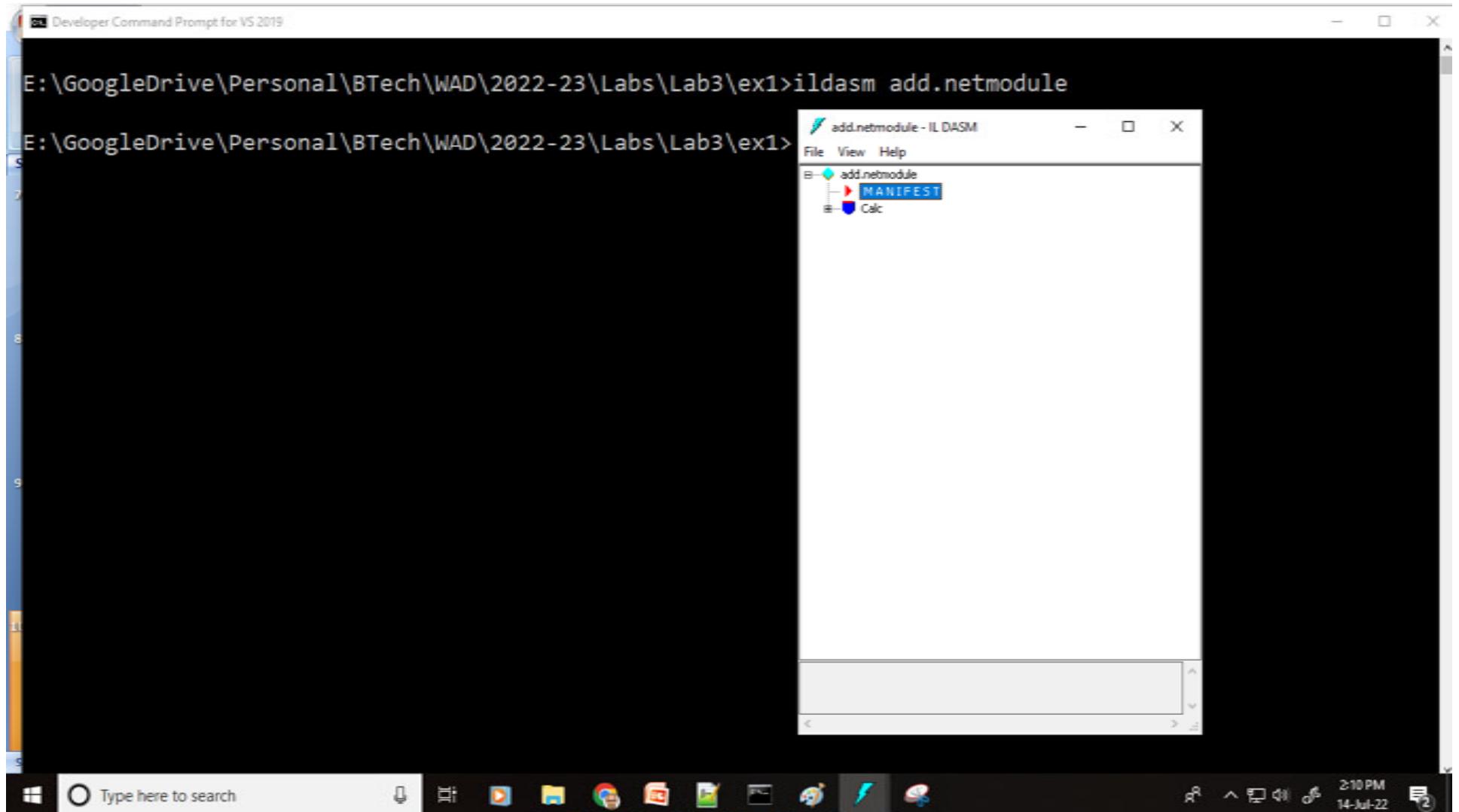
MANIFEST

Find Find Next

```
// Metadata version: v4.0.30319
.assembly extern mscorlib
{
    .publickeytoken = (87 7A 5C 56 19 34 E8 89 ) // .z\0.4...
    .ver 4:0:0:0
}
.assembly calc
{
    .hash algorithm 0x00008004
    .ver 0:0:0:0
}
.file add.netmodule
    .hash = (CA 4A 38 4D 43 7A 6F 17 27 9F F3 AC 4D E9 F1 B2 // .J8NCzo."...H...
            B9 F8 58 1B ) // ..X.
.file sub.netmodule
    .hash = (34 E8 08 D2 7A 81 2C BF B2 82 81 A2 28 BA 00 23 // 4...z.,..... ..#
            B3 7C 1B FF ) // -|..
.class extern public Calc.Add
{
    .file add.netmodule
    .class 0x02000002
}
.class extern public Calc.Sub
{
    .file sub.netmodule
    .class 0x02000002
}
.module calc.dll
// GUID: {136034a9-007a-47a3-9430-5077077283c6}
.imagebase 0x10000000
.file alignment 0x00000200
.stackreserve 0x00100000
.subsystem 0x0003 // WINDOWS_CUI
.corflags 0x00000001 // ILONLY
// Image base: 0x00CE0000
```

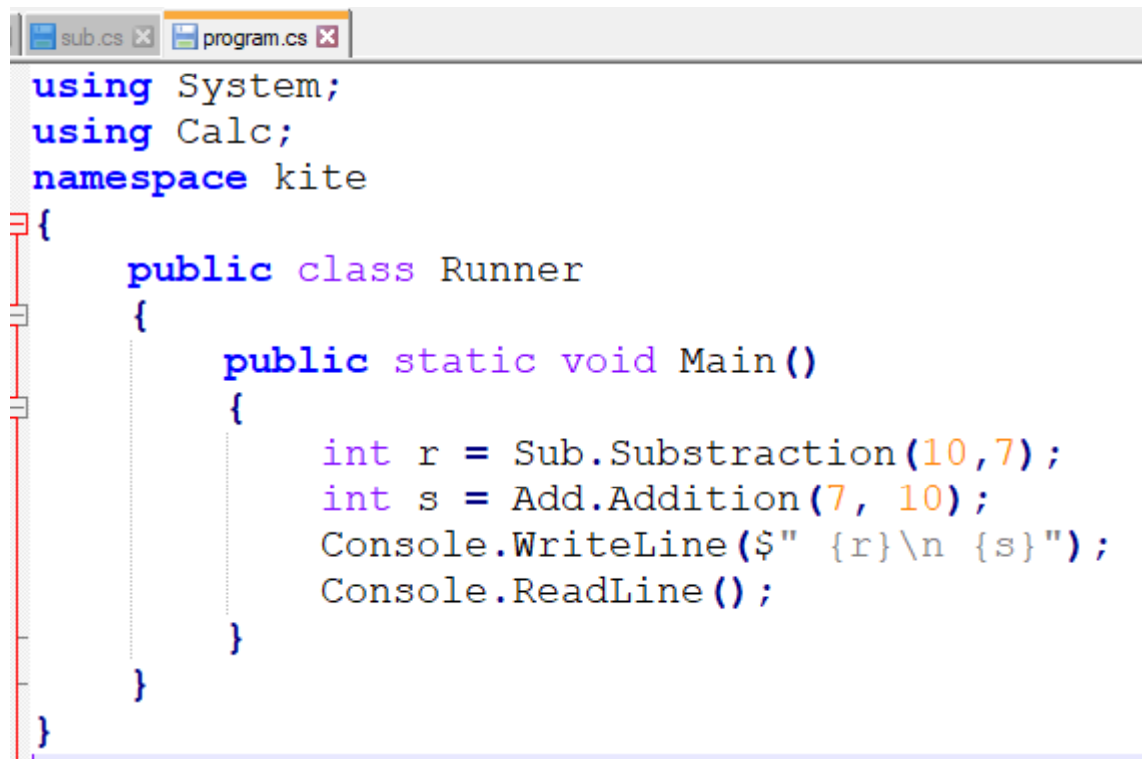
Type here to search

2:09 PM
14-Jul-22



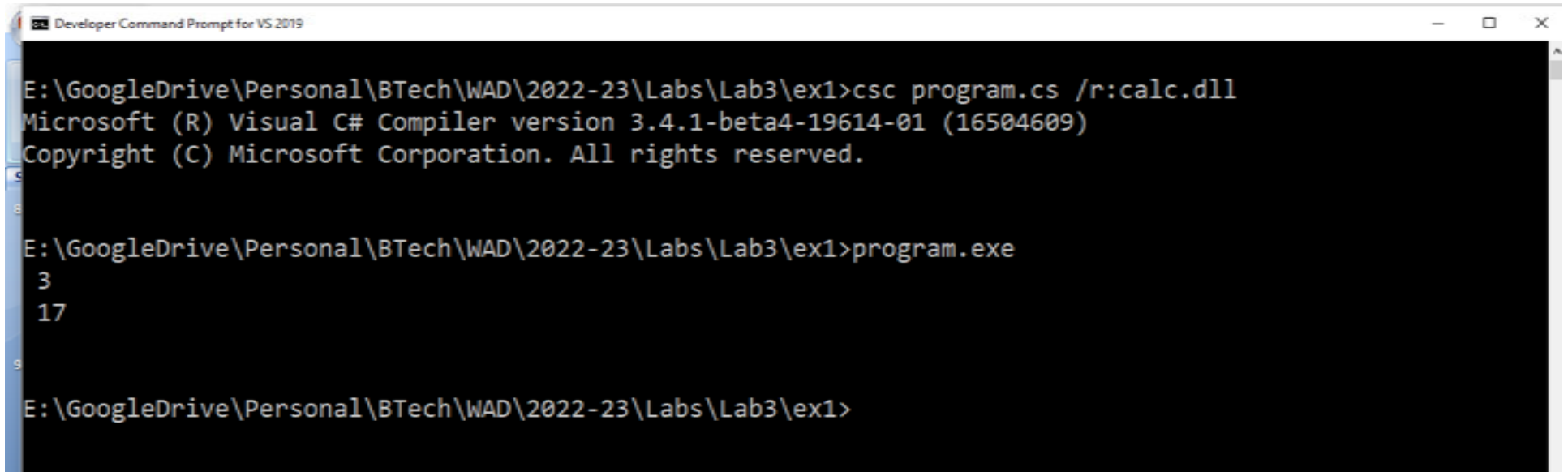
Here is the IL+ metadata. Why does it contain manifest?

Ex. 1



```
sub.cs x program.cs x
using System;
using Calc;
namespace kite
{
    public class Runner
    {
        public static void Main()
        {
            int r = Sub.Substraction(10,7);
            int s = Add.Addition(7, 10);
            Console.WriteLine($" {r}\n {s}");
            Console.ReadLine();
        }
    }
}
```

Ex. 1



```
Developer Command Prompt for VS 2019

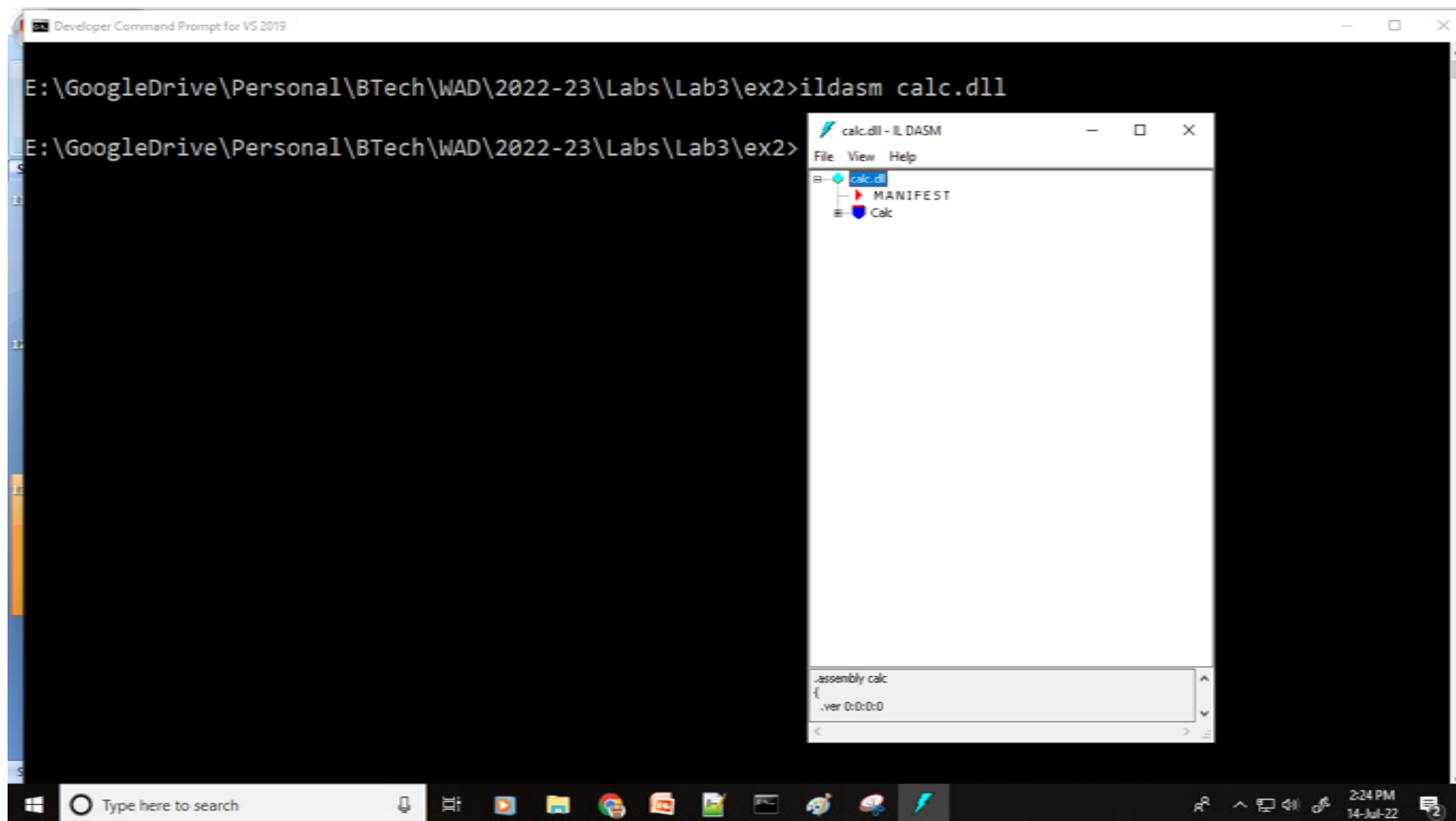
E:\GoogleDrive\Personal\BTech\WAD\2022-23\Labs\Lab3\ex1>csc program.cs /r:calc.dll
Microsoft (R) Visual C# Compiler version 3.4.1-beta4-19614-01 (16504609)
Copyright (C) Microsoft Corporation. All rights reserved.

E:\GoogleDrive\Personal\BTech\WAD\2022-23\Labs\Lab3\ex1>program.exe
3
17

E:\GoogleDrive\Personal\BTech\WAD\2022-23\Labs\Lab3\ex1>
```

Ex. 2

- Create a multifile assembly such that manifest is attached to one of the module.



Ex. 3

```
E:\GoogleDrive\Personal\BTech\WAD\2022-23\Labs\Lab3\ex2>SN -k MyCompany.snk  
Microsoft (R) .NET Framework Strong Name Utility  Version 4.0.30319.0  
Copyright (c) Microsoft Corporation.  All rights reserved.  
  
Failed to generate a strong name key pair -- Access is denied.
```

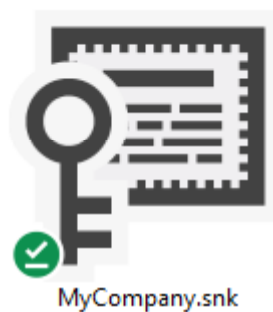
Ex. 3

```
Administrator: Developer Command Prompt for VS 2019
E:\GoogleDrive\Personal\BTech\WAD\2022-23\Labs\Lab3\ex3>SN -k MyCompany.snk

Microsoft (R) .NET Framework Strong Name Utility  Version 4.0.30319.0
Copyright (c) Microsoft Corporation.  All rights reserved.

Key pair written to MyCompany.snk

E:\GoogleDrive\Personal\BTech\WAD\2022-23\Labs\Lab3\ex3>
```



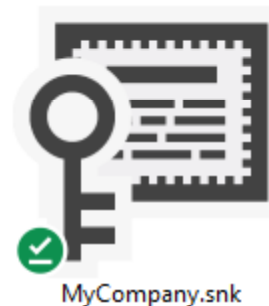
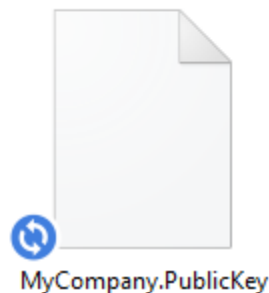
Ex. 3

```
Administrator: Developer Command Prompt for VS 2019
E:\GoogleDrive\Personal\BTech\WAD\2022-23\Labs\Lab3\ex3>SN -p MyCompany.snk MyC
ompany.PublicKey sha256

Microsoft (R) .NET Framework Strong Name Utility  Version 4.0.30319.0
Copyright (c) Microsoft Corporation.  All rights reserved.

Public key written to MyCompany.PublicKey

E:\GoogleDrive\Personal\BTech\WAD\2022-23\Labs\Lab3\ex3>
```



Ex. 3

```
Administrator: Developer Command Prompt for VS 2019
E:\GoogleDrive\Personal\BTech\WAD\2022-23\Labs\Lab3\ex3>SN -tp MyCompany.Public
Key

Microsoft (R) .NET Framework Strong Name Utility  Version 4.0.30319.0
Copyright (c) Microsoft Corporation.  All rights reserved.

Public key (hash algorithm: sha256):
002400000c800000940000000602000000240000525341310004000001000100051424f7365220
6ccd03f745843c9ca4f4aa5021a268fc0404e4c29efc1a6527bf6300126133ce2120012738df8e
7c1379d7cbd4ac3c0dcef3288c4c4a5ce3de0c9ccbc381ce7f0af93b1b97db49f7218810e73575
720b04610356a65ce5a720512f81fd29002ddaa454c3a40f4f0de43b03c8c3828bb7c7a8cf84ce
3dff56c3

Public key token is 2b1b98fa7fa14e30

E:\GoogleDrive\Personal\BTech\WAD\2022-23\Labs\Lab3\ex3>
```

Ex. 3

```
Administrator: Developer Command Prompt for VS 2019

E:\GoogleDrive\Personal\BTech\WAD\2022-23\Labs\Lab3\ex3>csc /keyfile:MyCompany.snk Program.cs /r:calc.dll
Microsoft (R) Visual C# Compiler version 3.4.1-beta4-19614-01 (16504609)
Copyright (C) Microsoft Corporation. All rights reserved.

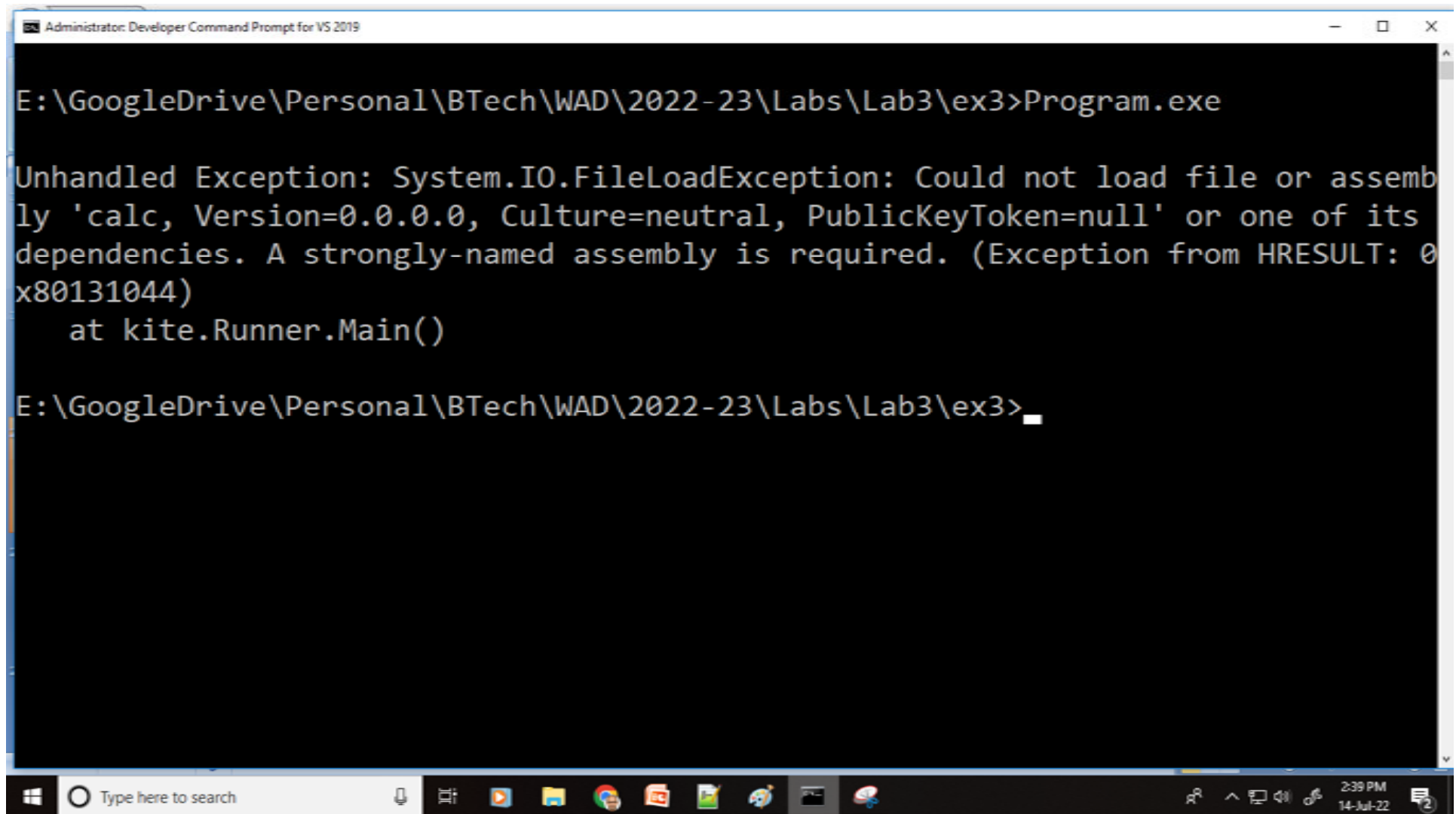
warning CS8002: Referenced assembly 'calc, Version=0.0.0.0, Culture=neutral, PublicKeyToken=null' does not have a strong name.

E:\GoogleDrive\Personal\BTech\WAD\2022-23\Labs\Lab3\ex3>_
```

```
Administrator: Developer Command Prompt for VS 2019

E:\GoogleDrive\Personal\BTech\WAD\2022-23\Labs\Lab3\ex3>Program.exe ?
```

Ex. 3



The screenshot shows a Windows Developer Command Prompt window titled "Administrator: Developer Command Prompt for VS 2019". The command prompt is at the directory `E:\GoogleDrive\Personal\BTech\WAD\2022-23\Labs\Lab3\ex3`. The user has executed the command `Program.exe`. The output shows an unhandled exception: `System.IO.FileLoadException: Could not load file or assembly 'calc, Version=0.0.0.0, Culture=neutral, PublicKeyToken=null' or one of its dependencies. A strongly-named assembly is required. (Exception from HRESULT: 0x80131044)`. The exception occurred at `kite.Runner.Main()`. The prompt is now waiting for the next command.

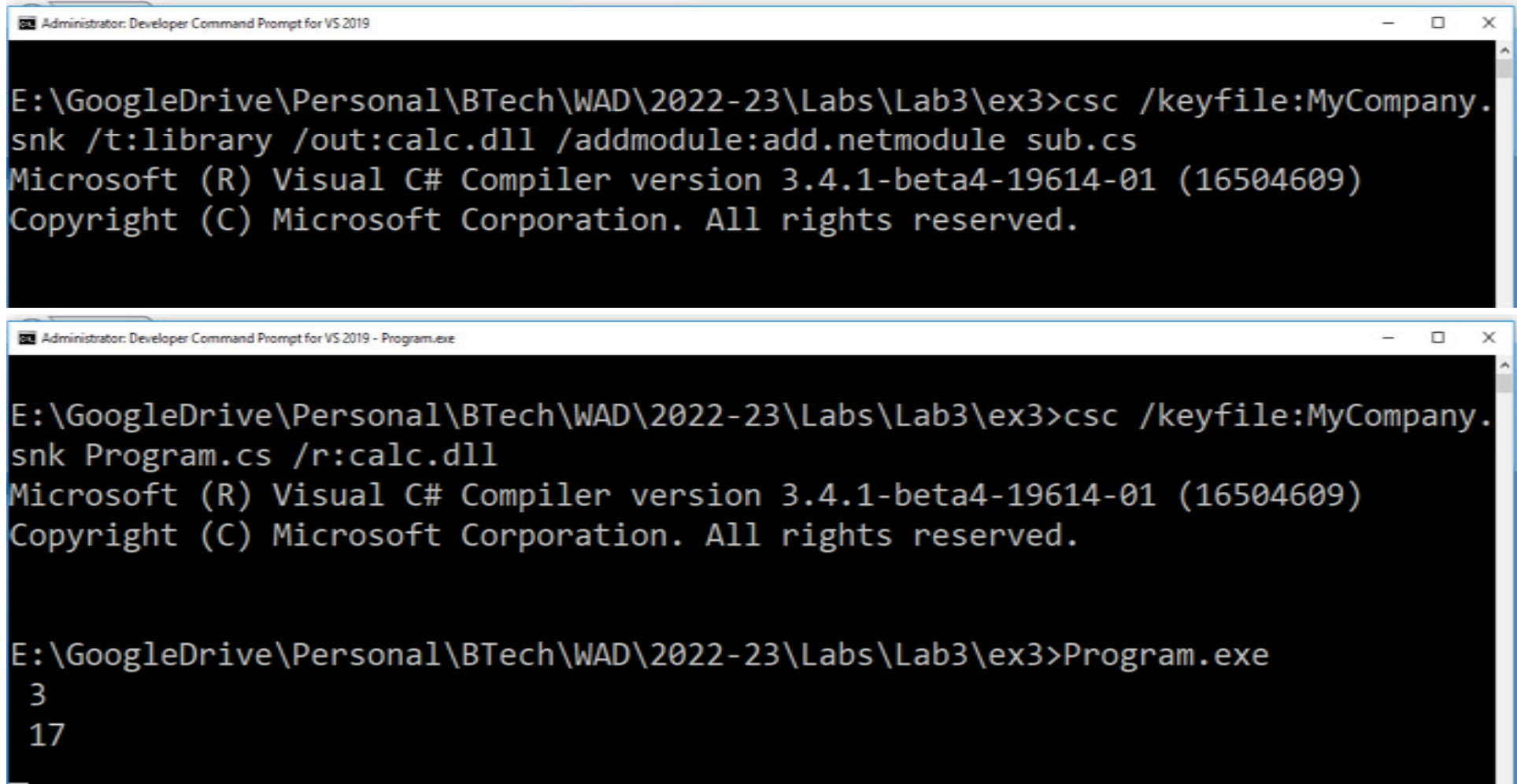
```
Administrator: Developer Command Prompt for VS 2019

E:\GoogleDrive\Personal\BTech\WAD\2022-23\Labs\Lab3\ex3>Program.exe

Unhandled Exception: System.IO.FileLoadException: Could not load file or assembly
ly 'calc, Version=0.0.0.0, Culture=neutral, PublicKeyToken=null' or one of its
dependencies. A strongly-named assembly is required. (Exception from HRESULT: 0
x80131044)
   at kite.Runner.Main()

E:\GoogleDrive\Personal\BTech\WAD\2022-23\Labs\Lab3\ex3>
```

Ex. 3



The image shows two screenshots of a Windows Command Prompt window. The top screenshot shows the compilation of a C# program using the Visual C# Compiler. The command is `csc /keyfile:MyCompany.snk /t:library /out:calc.dll /addmodule:add.netmodule sub.cs`. The output shows the compiler version 3.4.1-beta4-19614-01 (16504609) and the copyright notice for Microsoft Corporation. The bottom screenshot shows the execution of the program `Program.exe`. The output shows the number 3 on the first line and 17 on the second line.

```
Administrator: Developer Command Prompt for VS 2019
E:\GoogleDrive\Personal\BTech\WAD\2022-23\Labs\Lab3\ex3>csc /keyfile:MyCompany.snk /t:library /out:calc.dll /addmodule:add.netmodule sub.cs
Microsoft (R) Visual C# Compiler version 3.4.1-beta4-19614-01 (16504609)
Copyright (C) Microsoft Corporation. All rights reserved.

Administrator: Developer Command Prompt for VS 2019 - Program.exe
E:\GoogleDrive\Personal\BTech\WAD\2022-23\Labs\Lab3\ex3>csc /keyfile:MyCompany.snk Program.cs /r:calc.dll
Microsoft (R) Visual C# Compiler version 3.4.1-beta4-19614-01 (16504609)
Copyright (C) Microsoft Corporation. All rights reserved.

E:\GoogleDrive\Personal\BTech\WAD\2022-23\Labs\Lab3\ex3>Program.exe
3
17
```

How can we verify that assemblies are strongly named?